Causal Effect of Job Training on Earnings

Causal Question

Does participation in a job training program cause higher post-program earnings?

Identification Strategy

Double Machine Learning (DML)

Key Assumption

Conditional independence: after controlling for observed covariates, treatment assignment is independent of potential outcomes.

Why Prediction is Insufficient

Prediction models capture correlations but cannot isolate causal effects. Because program participation is not random, a predictive model would confound treatment effects with pre-existing differences.

Dataset

Lalonde job training dataset (MatchIt via statsmodels), N = 614

Treatment: treat, Outcome: re78, Controls: demographic variables and prior earnings (re74, re75)

Although the sample size is below 1,000 (N = 614), the Lalonde dataset is a standard benchmark in causal inference and is widely used in both academic research and applied econometrics. Moreover, it is explicitly listed under the “Lalonde variants” option in the project guidelines, making it an appropriate and valid dataset choice for this analysis.

Preliminary Findings

Treated individuals have lower pre-treatment earnings, indicating selection bias. The naive estimate is negative (−635), while the DML estimate is positive (~1245), suggesting that the naive estimate is downward biased.

Causal Estimation

DML is used to estimate the treatment effect while flexibly controlling for observed confounders.

Reproducibility

Code is maintained in a GitHub repository with version control and issue tracking.